

Prathyusha keerthi

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Education

Old Dominion University, MS in Computer Science

Aug 2023 – May 2025

- GPA: 3.8/4.0
- **Coursework:** Computer Architecture, Data Analytics, Information Assurance-Cybersecurity, Digital Image processing, Foundations of Computing

KL University, India, Undergrad in Computer Science

June 2018 – May 2022

- GPA: 8.4/10
- **Coursework:** Web programming, Database Concepts, Data Science, Data Visualization, Informatica Cognos

Experience

Full-Stack Developer Intern, Old Dominion University – Norfolk, Virginia

Jan 2024– May 2024

- Contributed to building a responsive and intuitive user interface using React.js, improving usability for students and faculty. Assisted in developing backend services with Node.js and PostgreSQL, helping automate scheduling, role-based access, and student record management.
- Built interactive and reusable UI components for different user roles, improving page performance and user experience. Helped implement backend features and RESTful APIs to support notification workflows and streamline communication across the platform. Collaborated in an Agile team environment, participated in code reviews, and used Git for version control and project collaboration.

Programmer Analyst, Cognizant – Hyderabad, India

Jan 2022– Aug 2023

- Developed and implemented robust ETL workflows using Informatica PowerCenter for enterprise-wide data integration, transformation, and migration across multiple source systems. Designed, built, and maintained interactive reports and dashboards in IBM Cognos, enabling real-time data visualization and supporting strategic decision-making across business units.
- Optimized ETL performance through query tuning, data validation, and process automation, ensuring high data quality and system efficiency. Collaborated with cross-functional teams to gather requirements, translate business needs into scalable data solutions, and provide ongoing production support to reduce downtime and improve system reliability. Created detailed documentation of data pipelines, reporting logic, and support procedures to streamline onboarding and knowledge transfer.

Software Development Intern, Techimax – Hyderabad, India

May 2021 – Dec 2021

- Engineered a full-stack sales forecasting application using React.js for building responsive, interactive data visualizations and Node.js for backend logic and API development. Designed and implemented RESTful APIs to deliver model predictions and processed datasets to the frontend, ensuring seamless integration and real-time data access.
- Applied time series forecasting models, including Seasonal Naive, ARIMA, and Linear Regression, to analyze and predict sales trends, enhancing forecasting accuracy and supporting strategic decision-making. Developed reusable React components for comparing model outputs and visualizing trends, improving clarity of insights for business stakeholders. Optimized backend performance for data processing and model execution, reducing latency and significantly improving user experience.

Projects

Car Rental Web Application

- Designed and developed a full-stack car rental web application using React for the frontend, Node.js with Express.js for the backend, and PostgreSQL for data management. The platform allows users to browse available cars, perform advanced filtering based on model, type, or availability, and view detailed vehicle information. Implemented secure user authentication, personalized user portals for managing bookings, and a real-time availability system. Integrated a seamless booking workflow with date-based reservations, and built RESTful

APIs to manage users, vehicles, and transactions. The system supports dynamic search, intuitive UI, and scalable backend architecture, delivering a smooth and responsive rental experience.

- Tech stack: React.js, node.js, JavaScript, Postgre SQL.

Music Streaming Web Application

- Developed a responsive music streaming web application using React, offering personalized user experiences through dynamic playlist creation, curated recommendations, and interactive browsing. Built backend services with Node.js and managed user data, playlists, and preferences using PostgreSQL, ensuring a seamless and scalable listening experience.
- Tech Stack: React.js, node.js, JavaScript, Postgre SQL

Movie Recommendation System

- Developed a personalized movie recommendation system using the ALS algorithm of PySpark and Spark MLlib, optimizing parameters for collaborative filtering with RMSE evaluation. Pre-processed data by handling missing values, normalizing ratings, and expanding multi-genre movies for granular analysis. Dynamically generated user-specific recommendations and incorporated feedback loops for continuous model improvement. Built a scalable architecture capable of handling large datasets and real-time interactions, ensuring deployment readiness.
- Tech Stack: PySpark, Spark MLlib, ALS model.

Water Quality Data Normalization and Analysis

- Designed and implemented a normalized database schema for water quality data, creating six interrelated tables to optimize data storage and querying efficiency. Executed SQL queries to answer complex business questions, such as monthly averages, extreme values, and data consistency across stations. Created an ERD diagram to represent the relationships between tables, ensured data integrity through normalization and validation.
- Tech Stack: MySQL, SQL Queries, ERD design.

Alzheimer's Disease and Health Aging

- Applied machine learning and statistical techniques to analyze trends, prevalence, and risk factors related to Alzheimer's disease. Geospatial analysis performed using GIS tools to map the distribution of diseases and identify regional patterns. Used time series analysis to evaluate the temporal evolution of Alzheimer's prevalence and its societal impact.
- Tech Stack: Python, Tableau, SQL database, GIS tools.

Google Food Review system

- Restaurant reviews analyzed to identify the popularity and quality of the dish using machine learning algorithms. Developed a Google extension for seamless website integration and data scraping. Provided data-driven insights to help customers make informed decisions.
- Tech Stack: React, Data Scraping, Python.

Technologies

Languages: C++, C, JAVA, JavaScript, TypeScript, SQL, JSON, HTML, CSS, Python, Tableau, Informatica, Matlab

Frontend: React, React Native, Angular, HTML5, CSS3, Material UI

Backend: Node.js, Express.js, RESTful APIs

Database: PostgreSQL, Firebase

AI/ML & Data: Pandas, NumPy

Visualization ETL: Tableau, Cognos, Informatica

Testing: JUnit, Postman